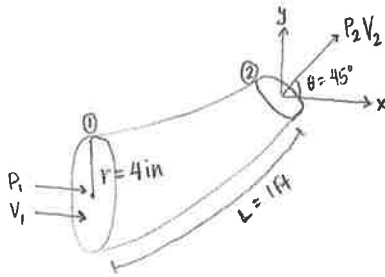


1)



10

$$\theta = 45^\circ \quad \sin(45^\circ) = \frac{\sqrt{2}}{2} \quad \cos(45^\circ) = \frac{\sqrt{2}}{2}$$

$$P_1 = 21 \text{ psi} = 21 \frac{\text{lb}_f}{\text{in}^2} \quad P_2 = 19 \text{ psi} = 19 \frac{\text{lb}_f}{\text{in}^2}$$

$$Q = 3 \frac{\text{ft}^3}{\text{s}}$$

$$P = 62.4 \frac{\text{lb}_m}{\text{ft}^3}$$

$$R = 4 \text{ in} = \frac{4}{12} \text{ ft} \quad D = 8 \text{ in} = \frac{8}{12} \text{ ft} = \frac{2}{3} \text{ ft}$$

$$L = 1 \text{ ft}$$

$$W = Q \cdot P$$

$$W = 3 \frac{\text{ft}^3}{\text{s}} \cdot 62.4 \frac{\text{lb}_m}{\text{ft}^3} = 187.2 \frac{\text{lb}_m}{\text{s}}$$

$$S_1 = S_2 = S$$

$$S = \frac{1}{4} \pi D^2$$

$$S = \frac{1}{4} \cdot \left(\frac{8}{12} \text{ ft}\right)^2 \pi = \frac{1}{9} \pi \text{ ft}^2$$

$$Q = \langle v \rangle \cdot S$$

$$V_1 = V_2 = V$$

$$V = S \cdot L$$

$$V = \frac{1}{9} \pi \text{ ft}^2 \cdot 1 \text{ ft} = \frac{1}{9} \pi \text{ ft}^3$$

$$F_{f \rightarrow s, x} = \frac{\langle v_1^2 \rangle}{\langle v_1 \rangle} w_1 \vec{u}_1 - \frac{\langle v_2^2 \rangle}{\langle v_2 \rangle} w_2 \vec{u}_2 + P_1 S_1 \vec{u}_1 - P_2 S_2 \vec{u}_2$$

$$= \frac{Q}{S} w (\vec{u}_1 - \vec{u}_2) + S (P_1 \vec{u}_1 - P_2 \vec{u}_2)$$

$$= \frac{3 \frac{\text{ft}^3}{\text{s}}}{32.174 \frac{\text{lb}_m \cdot \text{ft}}{\text{lb}_f \cdot \text{s}^2} \cdot \frac{1}{9} \pi \text{ ft}^2} \cdot 187.2 \frac{\text{lb}_m}{\text{s}} (1 - \cos(45^\circ)) + \frac{1}{9} \pi \text{ ft}^2 \left( 21 \frac{\text{lb}_f}{\text{in}^2} \cdot 1 - 19 \frac{\text{lb}_f}{\text{in}^2} \cdot \cos(45^\circ) \right) \cdot \left( \frac{12 \text{ in}}{1 \text{ ft}} \right)^2$$

$$= 394.90 \text{ lb}_f$$

$$F_{f \rightarrow s, y} = \frac{\langle v_1^2 \rangle}{\langle v_1 \rangle} w_1 \vec{u}_1 - \frac{\langle v_2^2 \rangle}{\langle v_2 \rangle} w_2 \vec{u}_2 + P_1 S_1 \vec{u}_1 - P_2 S_2 \vec{u}_2 - mg$$

$$= -\frac{Q}{S} w \vec{u}_2 - P_2 S_2 \vec{u}_2 - \rho g V$$

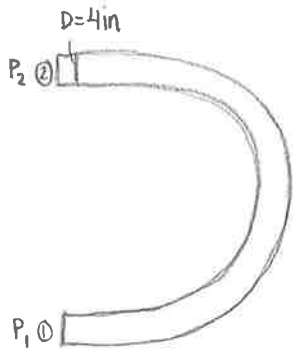
$$= -\frac{3 \frac{\text{ft}^3}{\text{s}}}{32.174 \frac{\text{lb}_m \cdot \text{ft}}{\text{lb}_f \cdot \text{s}^2} \cdot \frac{1}{9} \pi \text{ ft}^2} \cdot 187.2 \frac{\text{lb}_m}{\text{s}} \sin(45^\circ) - 19 \frac{\text{lb}_f}{\text{in}^2} \cdot \frac{1}{9} \pi \text{ ft}^2 \sin(45^\circ) \cdot \left( \frac{12 \text{ in}}{1 \text{ ft}} \right)^2 - 62.4 \frac{\text{lb}_m}{\text{ft}^3} \cdot \frac{1}{9} \pi \text{ ft}^3 \text{ lb}_f$$

$$= -732.46 \text{ lb}_f$$

$$F = \sqrt{(394.90)^2 + (-732.46)^2}$$

$$F = 832.13 \text{ lb}_f$$

2)



$$P_1 = 21 \text{ psi} = 21 \frac{\text{lb}_f}{\text{in}^2} \quad P_2 = 19 \text{ psi} = 19 \frac{\text{lb}_f}{\text{in}^2}$$

$$Q = 3 \frac{\text{ft}^3}{\text{s}}$$

$$\rho = 62.4 \frac{\text{lb}_m}{\text{ft}^3}$$

$$W = Q \cdot \rho$$

$$W = 3 \frac{\text{ft}^3}{\text{s}} \cdot 62.4 \frac{\text{lb}_m}{\text{ft}^3} = 187.2 \frac{\text{lb}_m}{\text{s}}$$

$$S_1 = S_2 = S$$

$$S = \frac{1}{4} \pi D^2$$

$$S = \frac{1}{4} \pi \left( \frac{4}{12} \text{ft} \right)^2 = \frac{1}{36} \pi \text{ft}^2$$

$$F_{f \rightarrow s, x} = \frac{\langle V_1^2 \rangle}{\langle V_1 \rangle} w_1 \vec{u}_1 - \frac{\langle V_2^2 \rangle}{\langle V_2 \rangle} w_2 \vec{u}_2 + P_1 S_1 \vec{u}_1 - P_2 S_2 \vec{u}_2$$

$$= \frac{Q}{S} w (u_1 - u_2) + S (P_1 u_1 - P_2 S_2)$$

$$= \frac{3 \frac{\text{ft}^3}{\text{s}}}{32.174 \frac{\text{lb}_m \cdot \text{ft}}{\text{lb}_f \cdot \text{s}^2} \cdot \frac{1}{36} \pi \text{ft}^2} \cdot 187.2 \frac{\text{lb}_m}{\text{s}} (1 - (-1)) + \frac{1}{36} \pi \text{ft}^2 \left( 21 \frac{\text{lb}_f}{\text{in}^2} \cdot (1) - 19 \frac{\text{lb}_f}{\text{in}^2} \cdot (-1) \right) \cdot \left( \frac{12 \text{in}}{1 \text{ft}} \right)^2$$

$$= 902.70 \text{ lb}_f$$